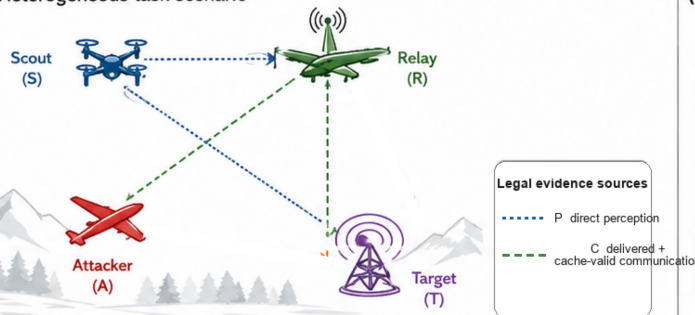
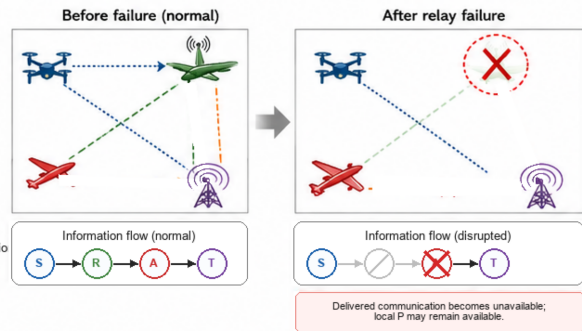


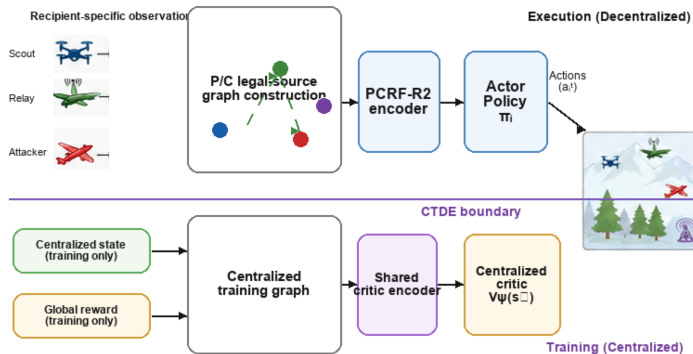
### (a) Heterogeneous task scenario



### (b) Failure disrupts coordination



### (c) Overall method pipeline (PCRF-R2)



### (d) PCRF-R2 encoder (zoom-in)

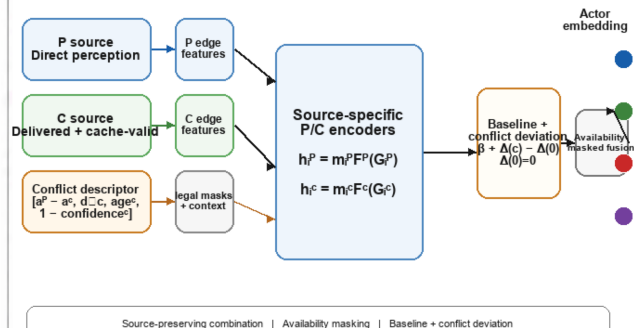


Fig. 1 | PCRF-R2: architecture overview for heterogeneous UAV coordination under relay failure and communication constraints.